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RESEARCH ARTICLE

Effects of scenery frame on visual depth perception in classical Chinese gardens: A case study of the Lvyin Pavilion in Lingering Garden

Haipeng Zhu [a], Yuhang Kong [b], Hong Zhang [a], Zongchao Gu [a,*], Ryuzo Ohno [c]

[a] *Department of Architecture, School of Architecture, Southeast University, Nanjing 210096, China*
[b] *Department of Architecture, School of Architecture, Tianjin University, Tianjin 300072, China*
[c] *Tokyo Institute of Technology, Tokyo, Japan*

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China

Abstract Visual depth (distance) perception is a fundamental aspect of environmental cognition, as it allows people to judge the spatial scale of their surroundings. However, estimating the depth of classical Chinese gardens is challenging, especially from static viewpoints that frame the scenery. Previous studies have examined how the internal components of the scenery frame affect depth perception. Still, the role of the frame and its peripheral information as environmental background have been largely overlooked. This study investigates how depth perception at viewpoints is influenced by viewing position displacement, frame geometry, and environmental context. The authors created nine stimulus materials in a cave virtual reality environment (three image treatments × three positions). Seventy-one participants were asked to evaluate depth perception using the magnitude estimation and adjustment methods. Their eye movement behavior was also recorded using an eye-movement instrument (SensoMotoric Instruments (SMI) eye-tracking glasses, 120 Hz). The results showed that participants could perceive spatial depth differences between viewing positions even when the internal viewpoint displacement was small; frame shape did not significantly affect depth perception and gaze behavior; and peripheral visual information of the frame enhanced depth perception significantly. Moreover, the form of the environmental background, especially the position of the scenery window, strongly guided the participants' gaze. These findings suggest that ambient visual information significantly impacts environmental experience, which landscape designers should consider.

* Corresponding author.
E-mail address: gzc991001@hotmail.com (Z. Gu).
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1. Introduction

Classical Chinese gardens comprise pavilions connected by multiple paths that serve as viewpoints (Tong, 1984; Pan, 1963). As people move along a path, their perception of the garden's size changes because spatial depth, a crucial parameter for measuring the spatial scale, is hard to perceive accurately (Gu, 2016; Pan, 1963; Zhu et al., 2021). Viewpoints are usually located in the central landscape area of the garden, where the ceiling and the side walls form a frame that captures the panorama and provides a vantage point for obtaining an overall impression of the garden (Tong, 2016). However, previous studies have also shown that the perception of spatial depth at the viewpoint is unclear. Feng et al. (2016) conducted a qualitative analysis of how the garden technique of obscuring the ground or water surface through the frame affects the perception of spatial depth during the sequential experience of entering the viewpoint. However, few quantitative studies investigate whether different viewing positions within a viewpoint affect depth perception differently.

The frame at the viewpoint often acts as a viewfinder, which limits the visual field and directs the attention to the inside of the frame. In contrast, the frame and its surrounding environment are often ignored as background information. Most of the existing studies examine how the properties of the internal elements of the frame, such as scenery layers and visual depth cues, influence depth perception (Wang and Li, 2018; Zhu et al., 2022). On the other hand, the shape of the frame and the surrounding visual information are treated as environmental context, and their possible impact on depth perception is overlooked.

With the advancement of physiological measurement techniques, such as eye trackers that can record eye movement behavior, which is a useful tool often used to understand spatial perception behavior (Li et al., 2020), it is possible to explore how people look at different parts of the scene and how that relates to their depth perception. Previous research (i.e., Wang and Li, 2018) analyzed eye movement behavior to explain how frames affect the willingness to explore in classical gardens. In this study, the authors use a combination of subjective perception evaluation and eye-tracking technology to examine how depth perception at different observation positions within the viewpoint is affected by three factors: viewing position displacement, frame geometry, and environmental context. The authors also investigate the relationship between eye movement behavior and depth perception.

This study aimed to provide a more comprehensive understanding of the characteristics of classical garden experiences by examining visual depth perception among observation positions from a viewpoint. To this end, a combination of the magnitude estimation and adjustment methods was used in a laboratory experiment wherein the eye-movement data of the participants were analyzed to understand their perceptual behavior. The authors observed that, while existing research has qualitatively analyzed the uncertainty in individuals' depth perception at landscape viewpoints, it fails to quantitatively examine and validate this phenomenon. Thus, the authors hypothesized that there are significant differences in spatial depth perception among sequential observation points within a given viewpoint (H1). Further, they observed that, despite examining the influence of scene composition within the view frame of a viewpoint on depth perception, existing studies have largely overlooked the effects of the view frame itself and its surrounding environmental context on depth perception and eye movement behavior. Thus, building on H1, this study investigated the environmental factors that contribute to variations in depth perception. The authors hypothesized that the perspective effect of a frame, caused by a change in viewing position, significantly influences people's depth perception and guides their gaze behavior (H2). Lastly, they hypothesized that the peripheral visual information of a frame significantly enhances people's depth perception and guides their gaze behavior (H3).

2. Theoretical framework

2.1. The influence of scenery frames on landscape perception and eye movement behavior

Frames are essential principles for arranging the basic elements of the landscape, such as line, shape, color, and texture. They are used to select and extract deliberate images from the environment (Starke and Simonds, 2013). The "frame" is a collection of views that can organize the landscape differently. Therefore, it is important to investigate how frames affect people's eye movement behavior and perception of the environment. Ohno et al. (2002) examined how the appearance of visual information at the edges of frames influenced eye movement behavior and subjective evaluation of people during viewpoint movement in three types of exit spaces. The experiment showed that the frame type controlled the order of environmental elements appearing in the visual field, affecting the landscape impression and eye movement tendency. Furthermore, eye-tracking studies of scenes with different depth-of-field levels within frames in classical Chinese gardens have shown that frames made the distribution of eye fixations more concentrated and increased the depth-of-field levels with longer fixation duration (Wang and Li, 2018).

2.2. The concept and measurements of visual depth perception

Depth perception, as discussed in this study, refers to egocentric distance perception, which is the subjective perceived distance of an observer to a target object

(Renner et al., 2013). Visual depth (distance) perception is a basic covariate for people to judge spatial scale, and it is closely related to direction (Lee, 1970) and time (Crompton, 2006) perception.

Visual depth perception is influenced by visual depth cues, which include pictorial, motion-induced, and structural cues. Since the experiment used panoramic images as stimulus material, this study focuses on pictorial cues. Previous studies have examined the effects of depth cues, such as occlusion, relative size, and perspective, on spatial depth perception of environmental elements inside the frame within classical garden viewpoints. These studies showed that the visual depth cue variables inside the frame caused by the change of observation position did not significantly affect the depth perception (Stevens, 1975). However, other studies have demonstrated that environmental context influences visual depth perception. For example, individuals judge the same distance differently in distinct environmental contexts, such as halls, corridors, and open lawn environments (Lappin et al., 2006) and indoor versus outdoor environments (Witt et al., 2007). Therefore, this study explores the potential effect of out-frame peripheral visual information on depth perception.

This study used two psychophysical methods to evaluate depth perception: the magnitude estimation and adjustment methods. The magnitude estimation method, which has been used for a range of perceptual measures like brightness, loudness, or tactility (Stevens, 1975), asks individuals to estimate the strength of a physical stimulus by assigning a value proportionate to the intensity of the stimulus they perceive. This method involves presenting the subject with a standard stimulus material and designating it with a specific value; the subject then assigns a value to other stimuli by comparing them to the standard stimulus. The adjustment method, used to study several perceptual metrics like brightness and loudness (Mollon, 1992), asks participants to adjust varied stimuli to match a constant or standard. For instance, using a standard visual stimulus of a particular intensity and requesting the observer to adjust the comparison stimulus to match the standard stimulus.

2.3. Focal and ambient visual information

Focal vision and ambient vision are parallel processing systems in visual perception that efficiently process a massive amount of visual information in the environment (Eisenberg and Zacks, 2016; Guo et al., 2024; Krukar et al., 2020). Focal visual information is the perceiver's consciously selected, slow processing of localized, specific information in a fixed direction of vision. Typically, focal visual objects consist of discontinuous and distinct components. Discriminating and perceiving the observed objects produces an understanding of the observed objects. Ambient visual information is acquired unconsciously by the perceiver from multiple radial sight lines converging on the observation point (visual radiation), integrating information from the global scope environment, and instantly processing visual information. Typically, ambient visual objects are continuous environmental surfaces. Frames from garden viewpoints are frequently used to intercept the landscape, drawing the viewer's attention to the scene within the frame. In contrast, the frame and its background are often perceived as ambient visual information.

The traditional methods of recording and analyzing visual information (such as local perspective and ordinary photos) are limited by the angle and field of view and focus more on the visual information of a specific range, ignoring the important role of ambient vision in landscape experience. With the development of panoramic vision technology, compared with traditional cameras, data acquisition devices such as panoramic cameras and stations can record complete visual information within the full field of view (360° horizontal, 360° vertical), avoiding the omission of peripheral visual information outside the frame by ordinary photos because of the limited angle and field of view. Simultaneously, panoramic images can be easily combined with head-mounted virtual reality devices and cave automatic virtual environments (CAVE), feasible alternatives for simulating real environments under laboratory conditions (Akiyama et al., 2018; Browning et al., 2020).

2.4. Application of virtual reality environment and eye-movement devices in landscape visual perception research

Virtual reality environments and eye-movement devices are widely used in landscape visual perception research because they can provide efficient and realistic methods for studying various aspects of landscape perception, such as visual preference (Noland et al., 2017), element identification (Zhou et al., 2023), and type characterization (Dupont et al., 2014). This study used the CAVE system to present garden scenarios as panoramic images. A CAVE system is an immersive virtual reality system that uses projection equipment to project experimental scenarios onto three to six walls of a cube-shaped room. A CAVE system can provide central and peripheral visual information compared to head-mounted VR. The experimental setup is described in detail in Section 3.2.

Eye movement indicators, such as the total number of gaze points, gaze frequency, total gaze duration, and average gaze duration, can reveal the cognitive processes of landscape perception (Deubel and Schneider, 1996; Hoffman and Subramaniam, 1995; Holmqvist et al., 2011; Williams and Castelhano, 2019). Among these indicators, identification, and measurement of gaze behavior are the most essential. Gaze behavior is a visual activity lasting over 100 ms (or 0.1 s). Gaze frequency is the number of gaze behaviors per unit of time (typically 1 s). When people observe their surroundings, they scan all parts of interest and form multiple gaze points, which are integrated into a comprehensive cognitive image of the brain. In this study, the authors used a heat map of gaze points to visualize the spatial distribution of fixation points and reveal the fundamental environmental cognition pattern.

3. Research methods

3.1. Stimulus materials

The Lingering Garden is a key case in the dynamic visual experience of the classical Chinese garden because of its

complex spatial transitions and changes. Several landscape studies in the 1950s and 1960s published in the Journal of Architecture focused on this garden as an example (Guo and Zhang, 1963; Pan, 1963; Peng, 1963). The Lvyin Pavilion is an important viewpoint in the Lingering Garden tour sequence. It is the first open viewpoint facing the main landscape area after visitors enter the garden through a winding, narrow, and closed corridor (Guo and Zhang, 1963; Lu, 2016). From this pavilion, the viewer can obtain the first impression of the entire garden landscape (Peng, 1963) (Fig. 1). The landscape frame, formed by the ceiling, walls, and handrails, limits the visual range of viewers facing the garden (Tong, 2016). Feng et al. (2016) suggested that people's judgment of the spatial depth of the garden at Lvyin Pavilion showed uncertain characteristics, as evidenced by the difference in depth perception between various observation positions and the deviation from the actual depth of the garden. Based on these qualitative findings, this study aimed to quantitatively examine the difference in depth perception among viewing positions at the Lvyin Pavilion.

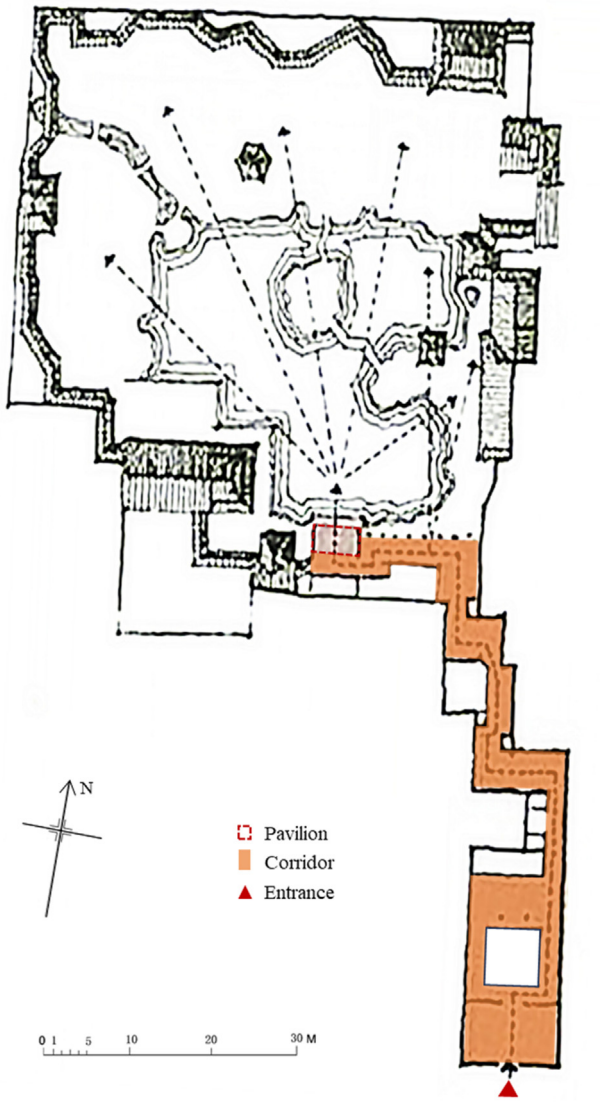


Fig. 1 Comparison of the spatial sequence from the entrance of Lingering Garden to the Lvyin Pavilion. Source: Peng (1963).

The location of the sampling points within the Lvyin Pavilion was primarily determined by three factors: the degree of visual information variation among the viewing positions, the geometry of the frame, and the frame's periphery. To obtain obvious changes in the frame shape and its peripheral visual information, three sampling points were selected within a small area of the pavilion (4.5 m × 3.5 m). Fig. 2 shows the left, middle, and right sampling points 1.5 m apart (corresponding to a normal step length of 75 cm for adults). Panoramic images were taken at a normal human visual height (1.6 m). The panoramic images were processed using Adobe Photoshop software to generate nine stimulus materials (three image treatments * three positions), as shown in Table 1. The image treatments were as follows: the interior space of the pavilion was darkened to eliminate the peripheral visual information of the frame (T1), the frame was treated as a rectangle to eliminate the influence of the frame shape (T2), and the original panoramic image was used to retain ambient visual information (T3).

3.2. Experimental equipment and methods

Participants assessed depth values using psychophysical methods (magnitude estimation and adjustment) in different experimental environments (CAVE and connected monitors). The participants wore SensoMotoric Instruments (SMI) eye-tracking glasses (120 Hz) throughout the experiment. The eye tracker vendor provided a software module to calibrate and record the frontal video frames and raw eye movement data.

3.2.1. Experimental equipment

In Test 1, participants experienced the stimulus material in CAVE. Panoramic images were projected onto three large projection walls (3.6 m × 2 m) to give the participants an immersive experience close to the real environment. Test 2 presented the panoramic images on three connected displays (0.57 m × 0.34 m). The display aspect ratio was 16:9, the same as the CAVE projection wall ratio used in Test 1. The experimenter avoided visual distortion after image stitching by adjusting the angle between the adjacent displays to ensure image continuity (Fig. 3).

3.2.2. Experimental method

The magnitude estimation method was used in Test 1, in which participants assigned values to other stimulus materials based on comparisons with the standard stimuli. The adjustment method was used in Test 2, in which participants adjusted the variable stimuli to match constant or standard stimuli. 3D modeling was used in Test 2 to simulate a scene with the corresponding viewing positions and perspectives. The ground grid (1.5 m × 1.5 m) in the 3D model was intended to provide depth cues to support distance judgments. The horizontal garden boundaries were fixed so that participants could modify the position of the target wall (i.e., the garden depth direction boundary) to match the perceived spatial depth in the panoramic image scene (Fig. 4).



Fig. 2 Plan of the observation positions at the Lvyin Pavilion.

3.3. Experimental design

The experiment consisted of two controlled experimental groups that examined the effects of frame shape and peripheral visual information on depth perception. Each group comprised 12 participants (Table 2). The overall design of the experiment is presented in Table 3.

3.3.1. Participants

Seventy-one participants (23 males and 48 females) were recruited via the Decision Science Lab (DeSciL) participant recruitment platform of ETH Zürich. The age of the participants ranged from 19 to 39 years ($M = 23.8$; $SD = 4.1$). Written informed consent was obtained from all participants prior to starting the experiment, in accordance with ethical guidelines. Participants were compensated 15 CHF per half hour and informed that they could withdraw from the experiment at any time.

Participants were required to have normal or corrected-to-normal vision (using contact lenses) to ensure accurate experimental results. Visual depth perception necessitates normal vision, and eye-tracking studies were conducted during the experiment. Therefore, participants needing vision correction were instructed to use contact lenses to avoid interference with the eye-tracking glasses.

None of the participants had visited the study site garden or had professional experience in architecture, landscape design, or related fields. Given that the experimental instructions involved judging the perceived spatial depth of the garden, the presence of participants with relevant professional backgrounds or actual touring experiences could influence the accuracy of visual depth perception judgments.

This study, which is a series of experiments in the literature (Zhu et al., 2022), involved 71 participants who evaluated the depth value of scenario A in the stimulus group. Among them, 12 participants also evaluated scenarios B and C. The depth values and eye-tracking data for this study were obtained from the corresponding participants.

3.3.2. Experimental process

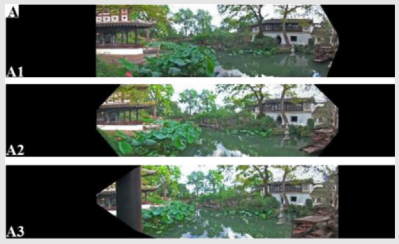
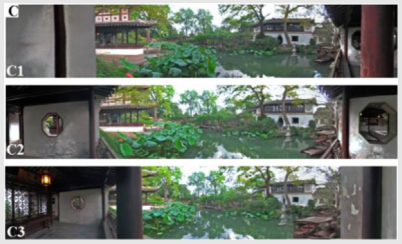
Test 1: Evaluating depth perception using the magnitude estimation method in a CAVE environment.

Step 1 Participants wore and calibrated the eye movement device.

Step 2 Test trials were conducted to familiarize the participants with the experimental procedure and test their understanding of the experimental instructions. The Daigo-ji Sanbo-in garden scene was selected to test the participants' depth estimation of the boundary wall (Fig. 5).

Step 3 Participants first viewed standard stimulus A2 with a depth value of 100 for 45 s. Then, participants viewed the comparison stimulus for 45 s and assigned a depth value to the comparison stimulus based on the impression of the standard stimulus. The comparison stimuli were tested in a randomized order. Before the next evaluation, the participants rested for 15 s with their eyes closed. For example, in the experiment contained stimulus groups A and B, where the standard stimulus was A2 = 100, and the comparison stimuli were A1, A3, B1, B2, and B3 for five depth evaluation trials (Fig. 6). Figure 7 shows the scene of one participant in Test 1.

Table 1 9 stimulus materials (3 image treatments × 3 observation positions).

Image treatments		Stimulus group	Observation positions (left, middle, right)
The shape of the frame	changed	A^{T1}	 (A1, A2, A3)
	unchanged	B^{T2}	 (B1, B2, B3)
Peripheral visual information of the frame	added	C^{T3}	 (C1, C2, C3)

Note: A^{T1}: Interior space of the pavilion was darkened for Group A; B^{T2}: The frame is treated as a rectangle for Group B; C^{T3}: The original panoramic image for Group C.

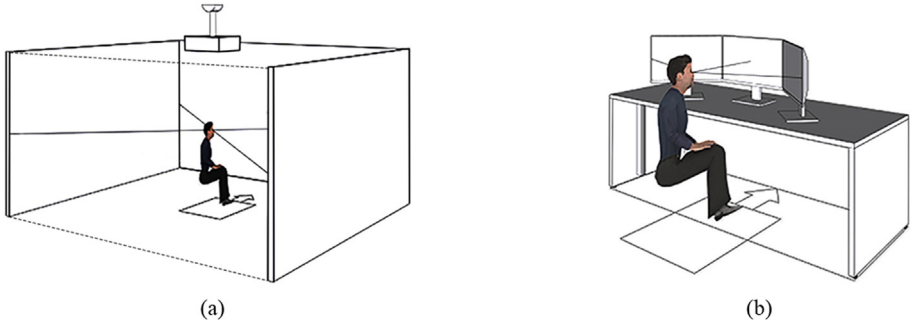


Fig. 3 Equipment of Tests 1 and 2. (a) Cave automatic virtual environment (CAVE); (b) three connected displays.

Test 2: Evaluating depth perception using the adjustment method in connected monitor.

Step 1 After the participant sat down, the investigator adjusted the seat height to align the participant's observation point with the center of the connected monitor.

Step 2 The participants completed a test trial to familiarize themselves with the experimental procedure and the specific operation of Test 2. The experimental instruction assessed the garden's spatial depth, that is, the perceived distance from the participant's position to the garden boundary wall. The garden depth in Test 2 was determined based on the location of the target wall in the 3D model. The starting position of the target wall in the 3D model was 200 m, whereas the actual distance of the garden was 60 m.

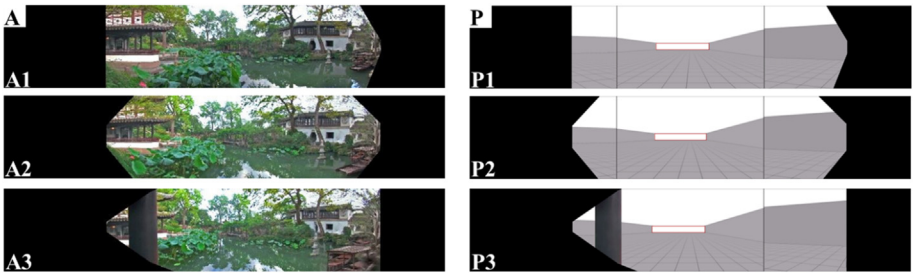


Fig. 4 Stimulus group P was the corresponding 3D model scene (P1: left, P2: middle, P3: right; the target wall was highlighted with a red frame).



Fig. 5 Panorama used in the test trail (cube projection).

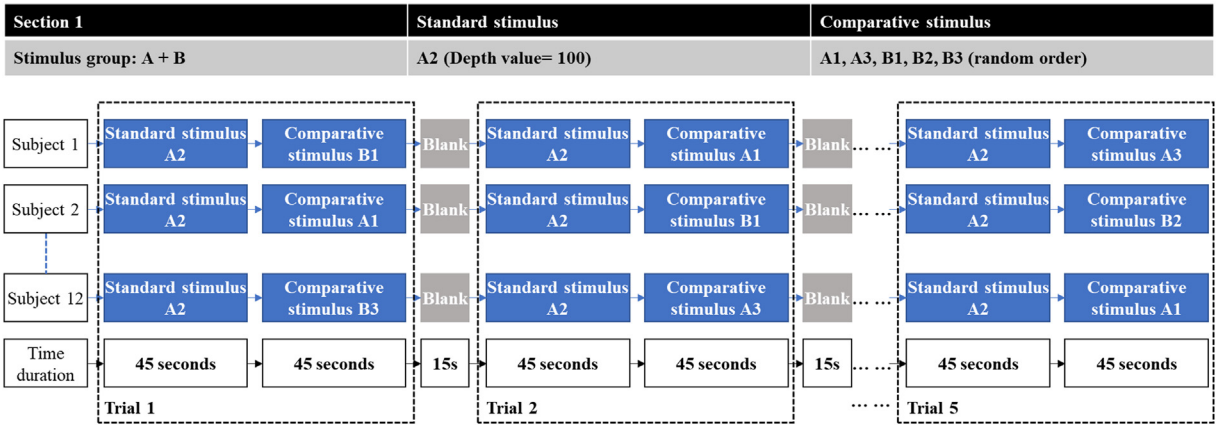


Fig. 6 Experimental procedure for Test 1 (example for stimulus groups A and B).

Step 3 The experiment consisted of six evaluation sets. In each test set, the participants were shown a panorama and its relative model (e.g., stimulus A1 and stimulus P1). Participants were instructed to use the "Alt + Esc" keys on the keyboard to switch between comparison stimuli and the "Up + Down" keys to modify the target wall in the 3D model. The participants could make as many changes as they wanted before matching the perceived depth of the panorama. The panoramas were randomly presented. A blank image (gray) was inserted between the panorama and the 3D model to prevent afterimage influence (Fig. 8).



Fig. 7 A student participating in Test 1.

4. Analysis and results

4.1. Data processing and visualization

Processing eye movement data involves converting raw data into comprehensible metrics and mapping them to correct panoramic image coordinates. This study used SMI BeGaze software to extract gaze position coordinate data in the observer's visual field coordinate system from the eye tracker data. It was output as video image data (24 frames per second). Owing to the frequent head rotation of



Fig. 8 A student participating in Test 2 (orange circles: gaze points; left: panorama A3; middle: blank image; right: corresponding model P3).

Table 2 Experiment containing two control experimental groups.

Section	Stimulus groups	*N*	Visual cues
1	A and B	12	shape of the frame
2	A and C	12	peripheral visual information of the frame

the observer during the experiment, it was not possible to directly obtain the position coordinates of the object under gaze in the image of the stimulus material. Therefore, the video frames were read individually using Visual Studio programming, and the gaze positions were manually annotated on the panoramic image coordinates. The program interface design is shown in Fig. 9.

4.2. Significant differences in depth perception among observation positions

This study obtained the results of 71 participants' evaluations of garden depth and excluded seven participants' data by outlier detection and incorrect manipulation during the experiment. Outlier detection of depth values was conducted using IBM SPSS 20.0. The criteria for experimental misoperation included misunderstanding of the concept of depth, failure to calibrate by the eye tracker, and overestimating depth values.

Table 4 shows the mean depth values ($N = 64$, $M = 23.41$, $SD = 3.61$) for different positions (stimulus group A) in Test 1. The depth values were evaluated by using the magnitude estimation method (ME values), with mean depth values of 92.86 ($SD = 19.11$) and 103.84 ($SD = 20.74$) for the left and right positions, respectively. The one-sample *t*-test showed a significant difference at the 0.01 level in the mean depth values between the left and middle positions and no significant difference between the right and middle positions. The paired sample *t*-test showed a significant difference at the 0.01 level between the left and right positions.

Table 5 shows the mean estimated distances ($N = 64$) obtained using the adjustment method for different viewpoint positions (stimulus group A) in Test 2. The mean estimated distances for the left, middle, and right positions were 119.44 m ($SD = 38.02$), 128.66 m ($SD = 42.38$), and 144.78 m ($SD = 37.52$), respectively. The paired sample *t*-test showed significant differences between depth values at different positions at the 0.01 level. The results of both Tests 1 and 2 indicated that there were significant differences in depth perception among the observation positions. Participants perceived the garden to be deeper when the position was sequentially changed from left to right.

The eye-movement data analysis in this study aims to illustrate and explain the environmental perceptual elements that influence the depth value results and demonstrate the view frame's effect on viewing behavior. The heat map of gaze points shows the areas with the longest gaze duration and the highest gaze frequency in yellow. The eye movement data of the first four valid subjects for each of the five experimental groups in the overall experimental design were selected, and twenty participants were analyzed.

Figure 10 shows the heat map of gaze points for 20 subjects viewing scenes from stimulus group A (A1, A2, and A3). In Tests 1 and 2, the most frequent and intense gaze locations were consistent with the experimental purpose of spatial depth judgment. This indicates that the subjects accurately comprehended the purpose of the experiment and actively collected, selected, and processed pertinent visual information to form a perception of the scene depth. Moreover, the visual behavior of the participants in Test 1

Table 3 The experiment containing two tests.

		Test 1	Test 2
Stimulus	Standard	A2	P (P1, P2, P3)
	Comparative	A (A1, A3) + B/C (B1, B2, B3/C1, C2, C3)	A (A1, A2, A3) + B/C (B1, B2, B3/C1, C2, C3)
Experimental setup		Cave automatic virtual environment	Three connected monitors
Eye movement collection equipment		Head-mounted eye tracker	Head-mounted eye tracker
Methods to obtain participants' response		Magnitude estimation method	Adjustment method
Subjective response		Depth value (ME value)	Estimated distance (m)



Fig. 9 Visual Studio coordinate matching interface.

Table 4 Mean depth values for the different positions in Test 1, **significant ($p < 0.01$).

Stimulus groups		*N*	Effect size (Cohen's *d*)	Power ($\alpha = 0.05$)	*p*	95% Confidence interval	
						Lower	Upper
A1	A2	64	0.37	0.83	0.004**	−11.91	−2.37
A2	A3	64	0.19	0.32	0.143	−1.34	9.02
A1	A3	64	0.45	0.94	0.007**	−18.90	−3.07

showed correlations with broader visual information. Therefore, the next section will discuss how frame shape and peripheral visual information affected the eye movement behavior of Test 1 participants.

The participants ($N = 20$) who judged the depth values of the three scenes of stimulus group A (A1, A2, A3) were ranked from highest to lowest, based on the absolute sum of the differences in depth values among the left, middle, and right positions. To compare their observation patterns, the first five participants were defined as the group with significant differences (Group 1), and the last five participants were defined as the group with insignificant differences (Group 2).

Figure 11 shows a heat map of gaze points for the participants (five each) with significant and insignificant differences in depth perception among the observation points. The two groups of participants differed in their choices of garden boundaries and visual focal objects. Group 1 focused on the pavilion on top of the hill, approximately 45 m from the planar distance of the observation point. Group 2 focused on the water source formed by the lotus and curved bridge, which was approximately 22 m from the planar distance of the observation point. Moreover, Group 1 showed variation in their focal observation area among the three scenes, while Group 2 showed consistency. These results suggest that participants in Group 1 focused on the further environmental element (pavilion) and showed uncertainty in the choice of garden boundary, which led them to perceive a significant difference in depth perception among observation points.

4.3. Effects of frame shape and its peripheral visual information on depth perception and gaze behavior

This section examines how the frame shape and its peripheral visual information affected the depth perception and gaze behavior of the participants in Test 1 using the magnitude estimation method (ME values). Table 6 lists the mean depth values of the scenes at different positions for the two experimental groups. The second experimental group had two stimulus groups, A and C, which differed in frame shape and peripheral visual information, respectively. The mean depth value for stimulus group A was 102.50 ($SD = 6.26$), and the mean value for stimulus group C was 116.39 ($SD = 19.88$). This shows that adding

Table 5 Mean depth values for the different positions in Test 2, **significant ($p < 0.01$).

Stimulus groups		*N*	Effect size (Cohen's *d*)	Power ($\alpha = 0.05$)	*p*	95% Confidence interval	
						Lower	Upper
A1	A2	64	0.34	0.76	0.009**	−16.01	−2.42
A2	A3	64	0.54	0.99	0.000**	−23.58	−8.67
A1	A3	64	0.81	> 0.99	0.000**	−33.17	−17.52

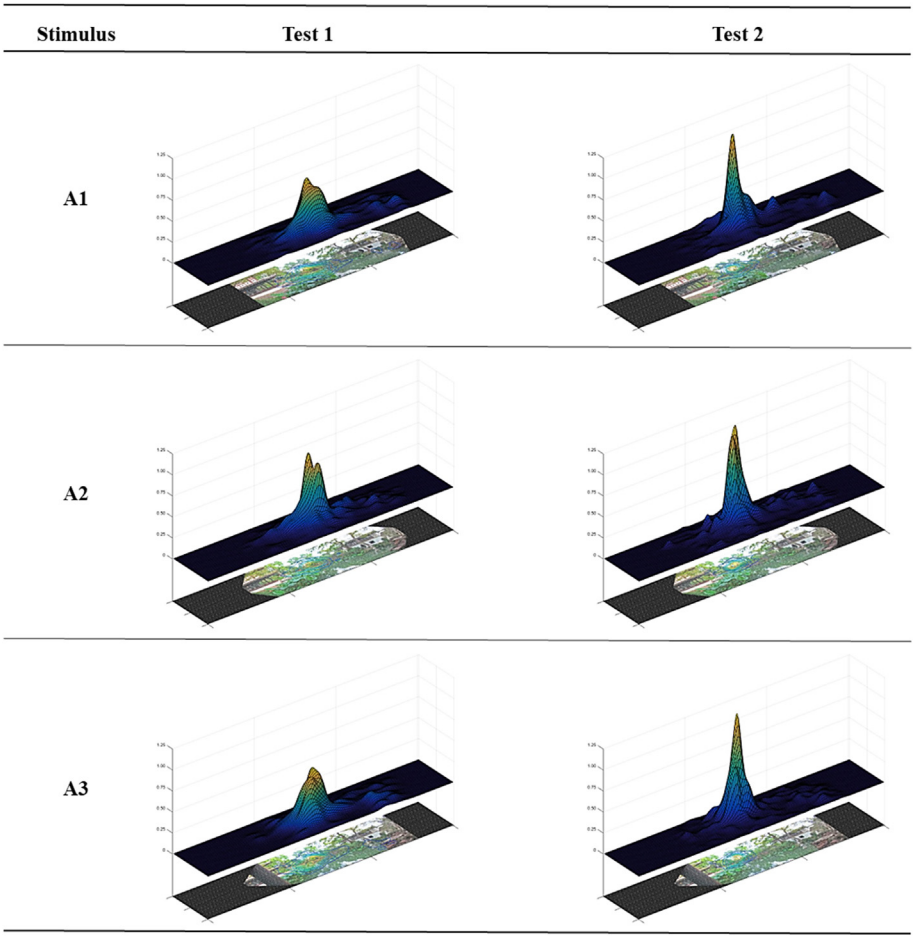


Fig. 10 Heat map of stimulus group A scenes in Test 1 and 2 (vertical coordinates represent the distribution of the Kanel density on a 360 × 180 grid).

peripheral visual information to the frame increased people's depth perception compared to the original stimuli. A paired sample *t*-test revealed a significant difference between the mean depth values of stimulus Groups A and C (t (11) = −2.798, p = 0.017, d = 0.942). This indicates that ambient visual information (i.e., the interior space of the Lvyin pavilion that formed the frame) significantly influenced depth perception.

This section examines how the frame shape and its peripheral visual information affected the depth perception and gaze behavior of the participants in Test 2 using the adjustment method. Table 7 lists the mean depth values of the scenes at different positions for the two experimental groups. In both the experimental groups, the estimated distances increased to various extents. A paired sample *t*-test showed a significant difference between the mean depth values of stimulus Groups A and C (t (11) = −2.311, p = 0.041, d = 0.27). This indicates that Test 2 further confirms the effect of the external context of the frame on depth perception.

In this study, the authors used heat maps of gaze points to explain the potential impact of a person's tendency to observe cognitive outcomes. Previous environmental behavior studies have attempted to corroborate and explain subjective perception results by selecting representative eye movement data for specific scenarios (Ohno et al., 2002). In this study, the authors chose the eye movement results of typical cases for different stimulus group scenarios to explain the differences in depth values between groups.

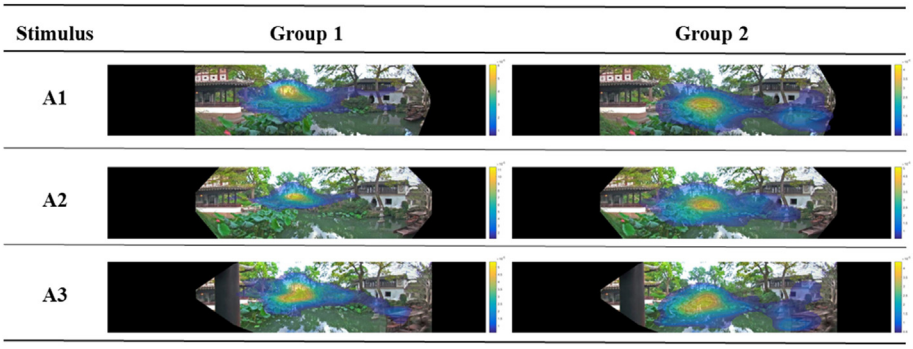


Fig. 11 Heat map of gaze points for participants (5 each) with significant and insignificant differences in perceived depth of scene A in stimulus Group 1 of Test 1.

Figure 12 shows the heat maps of gaze points for stimulus Groups A, B, and C scenes in Test 1. When comparing stimulus Groups A and B, there was no significant difference in the distribution of gaze points distributed along the water boundary. However, in scene A3, the highest gaze area was shifted to the left water boundary, compared to scene B3. This could be related to the shape of the frame in A3, which guides gaze behavior. In Stimulus group C (increasing the peripheral visual information of the frame), the participant's visual attention was guided by the indoor perspective of the Lvyin Pavilion while focusing on the garden boundary. In scenes C1 and C3, the indoor perspective changed the highest gaze area to the right and left sides of the frame, respectively, and the scenery windows in the interior enhanced the effect of the visual guidance.

The eye-movement results showed that the change in frame shape had no significant effect on gaze behavior. However, the peripheral visual information of the frame strongly guided gaze behavior, and the scenery windows in the frame's background environment played an important role in guiding visual attention.

5. Discussion

5.1. Significance

This study has both theoretical and practical significance. At the theoretical level, this study revealed the important influence of ambient visual information on the environmental experience. This showed that peripheral vision positively affected spatial depth perception compared to central visual information. At the practical level, this study examined how a rich, diverse, and substantial garden experience can be created in a limited green space. This study experimentally verified the effect of the gardening technique known as "hiding the source of water" on the perception of spatial scale in classical gardens. The implication of this finding for landscape designers is to use scenic objects (e.g., lotus, curved bridges, rockery) to obscure the boundary and source of the water surface. During observation from moving viewpoints, this occlusion causes the boundary of the water surface to change constantly, preventing observers from accurately gauging its size.

Table 6 Mean depth values of the scenes at different positions for the two experimental groups assessed using the measure estimation method (ME values) in Test 1; *significance ($p < 0.05$).

Stimulus groups		*N*	Mean ± *SD* (ME value)	Effect size (Cohen's *d*)	Power ($\alpha = 0.05$)	*p*	95% Confidence interval	
							Lower	Upper
1	A	12	97.89 ± 8.48	0.150	0.076	0.663	−11.17	7.39
	B		99.78 ± 15.67					
2	A	12	102.50 ± 6.26	0.942	0.843	0.017*	−24.81	−2.96
	C		116.39 ± 19.88					

Table 7 Mean depth values of the scenes at different positions for the two experimental groups assessed using the adjustment method (m) in Test 2; *significance ($p < 0.05$).

Stimulus groups		*N*	Mean ± SD (m)	Effect size (Cohen's *d*)	Power ($\alpha = 0.05$)	*p*	95% Confidence interval	
							Lower	Upper
1	A	12	131.83 ± 37.43	0.174	0.086	0.286	−18.61	6.05
	B		138.11 ± 34.54					
2	A	12	142.50 ± 35.24	0.270	0.137	0.041*	−19.09	−0.46
	C		152.28 ± 37.08					

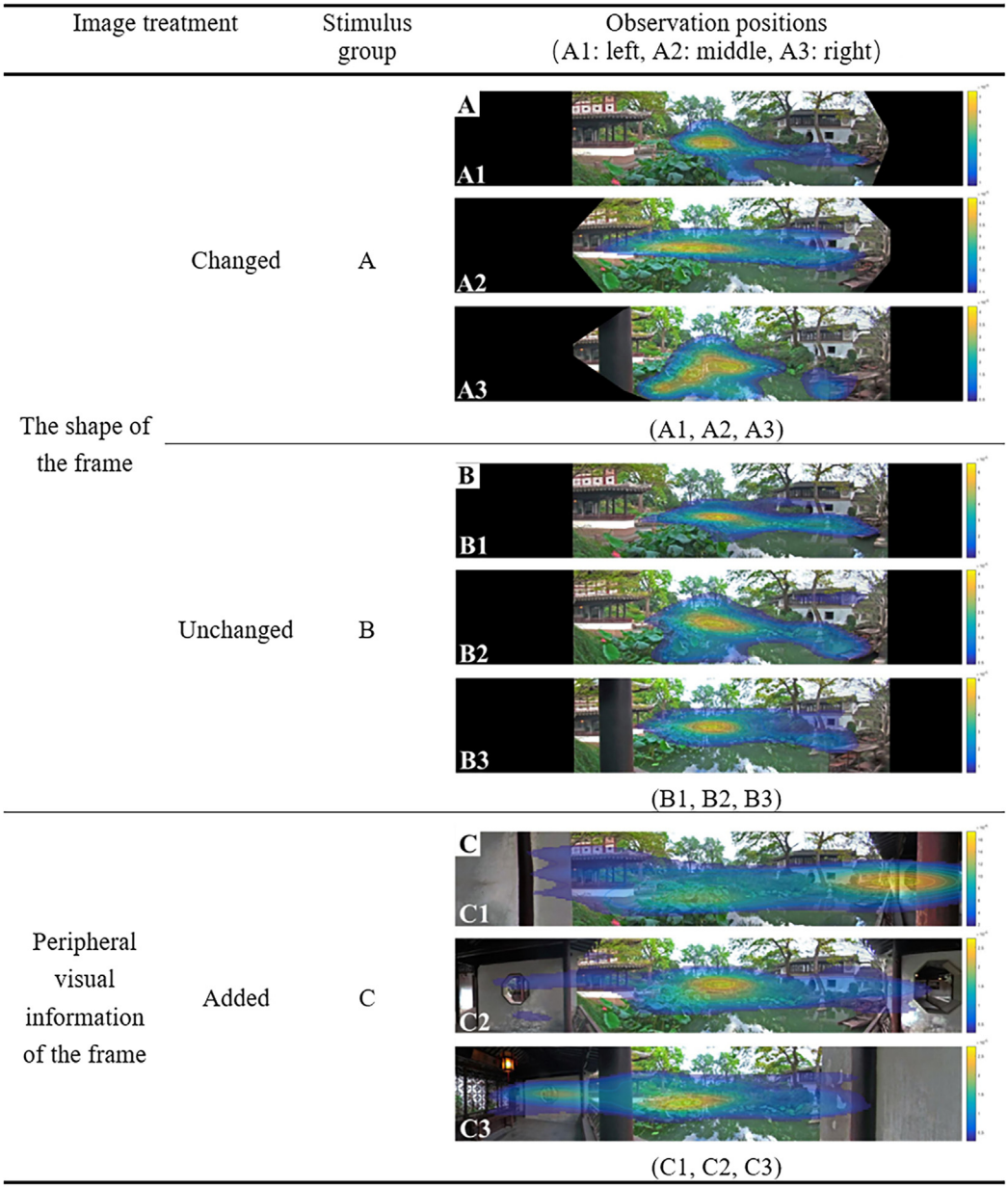


Fig. 12 Heat map of stimulus groups A, B, and C scenes in Test 1 (participant B-1, C-1).

Additionally, this study confirmed that setting up multiple potential visual attractors directs people's visual attention, thereby influencing their judgment of garden depth. This insight suggests that designers should incorporate multiple focal visual objects, such as water features, pavilions, and water sources, along the tour route. As viewpoints shift, these objects alternately become the visual focus, creating a rich visual experience and continually altering the basis for depth perception. Lastly, this study examined the impact of environmental context design on observers' subjective responses and visual behavior. Based on this finding, landscape designers should not only focus on the composition within the frame but also enhance the environmental background. Techniques such as setting up view windows can guide visual attention and enrich the overall visual experience.

5.2. Limitation

This study had some limitations and areas for improvement. First, it analyzed only the Lvyin pavilion of Lingering Garden as a case study to illustrate how the dynamic features of the gardening experience are manifested in its viewpoints. Future research should include multiple viewpoints within the same main landscape area of a garden and compare different garden viewpoints to draw more general conclusions regarding how frame elements affect depth perception. Second, the authors examined the effects of frame shape and peripheral visual information on depth perception. However, the sample size was limited. Subsequent studies should enhance research design and sample control. Although a larger sample size would increase the reliability of the results, the data

supporting the conclusions of this study have a statistical validity of more than 80%, which indicates that the experimental results have practical significance. Third, this study focused solely on the impact of the attribute of view framing on spatial depth perception, lacking a comprehensive examination of the correlation between garden design techniques and human spatial perception. Future research should conduct a series of cognitive experiments to systematically investigate visual perception in landscape design. This approach will help establish theoretical connections with landscape architecture and architecture, ultimately providing a more scientific basis for guiding practical applications in garden design.

6. Conclusion

This was a preliminary study on how the depth perception of observation points differs from that of classical garden viewpoints. The study explored how frame shape and peripheral visual information affect depth perception and gaze behavior. The results showed that: (1) the spatial depth perceived by the participants was significantly different even with a small change in the observation point's displacement inside the viewpoint; (2) the change in frame shape due to the change in observation point did not significantly affect depth perception or guide gaze behavior; (3) surprisingly, peripheral visual information had a significant enhancing effect on garden depth perception, although in the experimental task of judging the depth of the garden, that is, the location of the garden boundary wall in front, participants' gaze behavior was strongly influenced by the environmental context, such as the location of the scenery window, an environmental factor that was not considered in the initial experimental design.

The difference in participants' gaze behavior partly reflects their preferences for specific scenarios and can help explain and support differences in subjective evaluation results. Based on the spatial distribution of participants' gaze points, the environmental interface boundary is evidently one of the visual cues for people to judge garden depth. The uncertainty in choosing the garden boundary was the main reason for the differences in depth perception. The analysis of eye-movement data revealed perceptual elements related to depth perception, such as water surface boundaries, potential visual attraction points, environmental contexts, and the formal design of environmental backgrounds. The specific relevance of these perceptual elements to depth perception will be investigated in a future study.

Author contributions

H-PZ and RO: conceptualization, methodology. H-PZ and Z-CG: data collection and analysis. H-PZ: original draft. Y-HK and HZ: supervision. All authors contributed to manuscript revision, read, and approved the submitted version.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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